Shoumik Saha

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RESEARCH

- Computer Security
- Machine Learning
- Bioinformatics

EDUCATION

Bangladesh University of Engineering and Technology (BUET)
BSc. *Mar 2016 – Feb 2021*
Computer Science and Engineering

- CGPA – *3.66/4.0*
- 3rd year GPA – *3.84/4.0*
- Thesis supervisor – Dr. Atif Hasan Rahman

Notre Dame College
Higher Secondary Certificate *Jul 2013 – Jun 2015*

- GPA – 5.00/5.00

PROFESSIONAL EXPERIENCE

Lecturer
Department of Computer Science and Engineering *Jul 2021 – Present*
United International University

Research Assistant
Data Science and Engineering Research Lab *Mar 2021 – Present*
BUET
Under supervision of
Dr. Mohammed Eunus Ali, Dr. Atif Hasan Rahman

RESEARCH EXPERIENCE

- **Shoumik Saha**, Sadia Afroz, Atif Rahman. **MALIGN: Adversarially Robust Malware Family Detection using Sequence Alignment**
  Status: Proceeded to the rebuttal period of IEEE S&P 2022 (arXiv) (pdf)
- Subangkar Karmaker, **Shoumik Saha**, Atif Rahman, Mohammed Eunus, Mehedy Masud. **Detecting Atrial Fibrillation from PPG Signal using Similarity Based Self-supervised Learning**
  Status: Ongoing

SELECTED ACADEMIC PROJECTS

- **Website Development for Flight and Hotel Room Reservation System**
  Developed a website where customers can book flight tickets, hotel rooms through card payment, and companies can update their page and manage bookings.

Tools: Django, SQLite
Github link

- **Software Development for Football Club Management**
  This application stores all information about the club (players, managers, games). Admin of the club can update, add and delete these information and generate graphs from statistics.
  Tools: Java, JavaFX, Oracle Database
  Github link
- **Micro-controller Project for Real-time Detection of Car Theft**
  Can detect a car in real time and send a warning message with gps location to the owner's mobile for exceeding speed limit or crossing a safety zone. Even the owner can get the gps location of the car with a single message.
  Tools: Arduino, GPS, GSM module, Sonar, LDR sensor
  Youtube video link
- **Multi-player Shooting Game**
  A second-year project, two players can connect from two different machines under the same network and play penalty shoot and save, where one will shoot and another will save.
  Tools: Java, JavaFX, Java Networking
  Github link
- **Tic-Tac-Toe**
  A first-year project, single player game with Mini-Max algorithm.
  Tools: C, iGraphics, Artificial Intelligence
  Github link
- **Miscellaneous**
  - Developing a C compiler using lexical analyzer and parser designing tools
  - Designing a 4-bit Computer Model using Atmel Studio, MIPS architecture
  - Simulating Mancala game in AI lab course
  - Implementing and modifying some functionalities of XV6 Operating System
  - Modifying some functionalities of Computer Network in NS2.

COURSES INSTRUCTED

- Discrete Mathematics
- Data Structure and Algorithm
- Operating Systems
- Digital Logic Design

ACTIVITIES

- General Secretary, BUET Photographic Society *2020-2021*
- Co-host & Co-organizer, BUET CSE Fest *2020*
- Participant, ICPC Dhaka Regional *2018*
- President, Notre Dame Nature Study Society *2014*
- Vice President, Gregorian Science Club *2012*

STANDARDIZED TEST SCORES

- *TOEFL:* 111/120 (Reading: 28, Listening: 28, Speaking: 27, Writing: 28)
- *GRE:* 321/340 (Quant: 167, Verbal: 154, AWA: 3.5)